Dalhousie University

Department of Psychology

Nineteenth Annual In-House Convention

29-30 April 1993

SCHEDULE

Thursday, April 29, 1993

9:30 - 9:45 Coffee

9:45 - 10:00 Opening Remarks—Donald Mitchell, Acting Chair

10:00 - 11:30 Session 1. Chair: Vin LoLordo

T1) R. Klein, T. Taylor & A. Kingstone
T2) J. Christie & R. Klein
T3) V. Corkum & C. Moore
T4) M. Yoon & M. Yoon

11:30 - 12:30 Posters (Lounge):

P1) K. Frame, J. Christie & R. Klein
P2) J.P. Hamm & P. A. McMullen
P3) C. Dywan, J. Byrne & J.F. Connolly
P4) H. Schellinck & P.A. McMullen
P5) P.C. Kind, C.J. Beaver & D.E. Mitchell

12:30 - 2:00 Lunch

2:00 - 3:00 Session 2. Chair: Sandra Wright

T5) C. Moore
T6) J. Barresi & R. Martin
T7) Y. Gidron

3:00 - 3:30 Afternoon Tea

3:30 - 4:45 Session 3. Chair: Chris Beaver

T8) P.A. McMullen, J.D. Fisk & S.J. Phillips
T9) N.A. Phillips & J. McGlone
T10) B.J. Losier, J. McGlone & S.E. Black

5:00 - ... Gather in the Lounge for Wine and Cheese

Friday, April 30, 1993

9:45 - 10:00 Coffee

10:00 - 11:30 Session 4. Chair: Elizabeth Coscia

T11) F. Dastur & R.E. Brown
T12) H. Piggins & B. Rusak
T13) D. Phillips
T14) S.R. Shaw

11:30 - 12:30 Posters (Lounge):

P6) J. Francis & K. Davidson
P7) K.A. Frame
P8) M. MacGregor & K. Davidson
P9) R. Ingenmey & P. Dunham
P10) P. Hall & K. Davidson

12:30 - 2:00 Lunch

2:00 - 3:00 Session 5. Chair: Valerie Corkum

T15) A. Stuart & D. Phillips
T16) B. Rusak & H. Abe
T17) G. Eskes

3:00 - 3:30 Afternoon Tea

3:30 - 4:45 Session 6. Chair: Jim Clark

T18) T. Fidler & V.M. LoLordo
T19) W.K. Honig
T20) K.A. Forbes & J.F. Connolly

4:45 - 5:00 Closing Remarks and Awards: Kelly Forbes (Graduate Representative)

5:00 - ... Dinner in the Lounge

T1. Raymond Klein, Tracy Taylor and Alan Kingstone

WHY DO FIXATION OFFSETS FACILITATE SACCADIC LATENCIES?

When the fixation point is removed 200 ms prior to the appearance of a visual target (gap condition), subjects can initiate saccades within 100 ms. This is so much faster than the typical (when the fixation remains present, overlap condition) saccadic reaction time (RT) that the label "express saccade" has become popular. We will describe some problems with this label and suggest that investigators focus their attention on those factors which act to reduce RT in the gap condition. Our research suggests a 2-component model of the gap effect: 1) motor preparation which can operate in any response modality and is dependent on the foreperiod between any warning signal and a subsequent target, and 2) saccadic disengagement which (normally) depends on removal of the fixation stimulus, operates specifically to reduce saccadic latencies, and may be mediated by the rostral pole of the superior colliculus.

T2. John Christie and Raymond Klein

CHRONOMETRIC ANALYSIS OF THE EFFECTS OF FAMILIARITY UPON VISUAL ORIENTING: DOES WHAT WE KNOW AFFECT WHAT WE NOTICE, AND IF SO HOW AND WHEN?

It is often assumed (c<f> Farah) that the system responsible for perception of the visual array is richly interconnected with both a spatial attention system and a system of stored representations. Contemplating this arrangement one of us (Christie) proposed that familiar items should exert a stronger pull on attention than novel ones, because the region in the visual array occupied by a familiar item will be more highly activated due to greater feedback from its stored representation. The relevant empirical literature, which appears to be full of contradictions, will be critically reviewed. The method we have developed to explore our proposal will be described, and the timecourse of visual orienting to familiar vs unfamiliar objects, as seen using our method, will be revealed.

T3. Valerie Corkum and Chris Moore

DEVELOPMENTAL EMERGENCE OF JOINT VISUAL ATTENTION

The present study examined joint visual attention in a conditioned head turn paradigm in 63 infants from 6 to 11 months of age. Results suggested three distinct patterns of response, which were developmentally ordered. Older infants tended to show spontaneous JVA, without the need for conditioning. Infants at an intermediate age showed no spontaneous JVA but did learn to match an adult's head turn given contingent feedback. Finally, a perseverative pattern was demonstrated by the youngest infants who tended to show either a majority of head turns in one direction or several series of turns in the same direction. These results suggest that the onset of joint attention, in line with general sensorimotor development, may be tied to infant's ability to respond flexibly to two separate spatial locations on the basis of different cues.

T4. Myong G. Yoon and Myonghae Yoon

THE DEVELOPMENT OF CHILDREN'S UNDERSTANDING OF FALSE BELIEFS

We studied the developmental sequence in the following aspects of children's understanding of false beliefs: Predictions of a protagonist's action, explanations of their own predictions, explicit attributions of the protagonist's thinking, explanations of their own attributions, and retroactive justifications for the protagonist's futile past action. The results show that a child's ability to make a correct prediction of a protagonist's action develops earlier than his/her abilities to explain the correct prediction, to impute the protagonist's mental states, and/or to justify the protagonist's futile past action: During 3y 1m - 3y 6m, 52 % of younger 3-year-olds made correct predictions of the protagonist's action. Most (95 %) of them, however, were unable to make an explicit attribution of false belief to the protagonist. They also showed difficulties in justifying the protagonist's futile action retroactively, in spite of their preceding successful predictions of the protagonist's action. During 3y 7m - 4y 0m, 38 % of older 3-year-olds acquired new abilities to explicitly attribute false belief to the protagonist, whereas their correct predictions (61%) changed insignificantly. Longitudinal studies show that children's correct predictions of action developed 4 - 7 months earlier than their explicit attributions of false beliefs to protagonists. This developmental trend suggests that a normal child's understanding of false beliefs develops gradually from (a) the 'ego-centric (Level 0) conceptual perspective taking' stage (characterized by 'incorrect prediction of action and inability to attribute false belief') to (b) the transitory intermediate stage of 'Level 1 conceptual perspective taking' ('correct prediction of action without being able to attribute false belief'), and then to (c) the stage of 'Level 2 conceptual perspective taking' ('attribution of false belief as well as correct prediction of action') during the formative period between 3 and 4 years of age.

T5. Chris Moore

DEVELOPMENTAL RELATIONSHIPS BETWEEN PRODUCTION AND COMPREHENSION OF MENTAL TERMS

Production of mental terms, including belief and desire terms, was studied in a sample of 14 mother-child pairs longitudinally when the children were 2;0, 3;0, and 4;0. In addition, children at 4;0 were tested for comprehension of the distinctions between the belief terms 'know', 'think', and 'guess' using a task designed to assess the understanding of the expression of relative certainty. Children's comprehension of belief terms at 4;0 was significantly correlated with mothers' use of belief terms at 2;0, children's use of desire terms at 2;0 and children's use of belief terms at 3;0. These results reflect individual developmental differences in the development of belief-desire reasoning and are consistent with a picture of maternal linguistic influence on their children's developing theory of mind.

T6. John Barresi and Raymond Martin

THE VIEW FROM ELSEWHERE: OR, EVERYTHING THAT YOU WANTED TO KNOW ABOUT YOUR SELF BUT WERE AFRAID TO ASK

Instead of engaging in experimental research, philosophers like to explore theoretical questions by imagining various sorts of possible worlds interestingly different from our own. Such imaginative worlds can sometimes be used to answer questions inaccessible to empirical study. One such question that we hope to explore using this method in a future paper is: How would our self-knowledge change if we had access to perceiving ourselves from both a first and third person perspective? The way to approach this question is to imagine that a person could fission into two identical individuals, as do some other biological species on our planet, and that this is a typical happening in the adult lives of individuals on this imagined world. What would self-knowledge amount to under circumstances when you had recently divided in two? In this talk our intuitions into this matter will be explored.

T7. Yori Gidron

THE PSYCHOSOCIAL AND PHYSICAL PREDICTORS OF ADOLESCENTS' RECOVERY FROM ORAL SURGERY

This study expanded medical models of prediction and outcome related to surgery in adolescents. Fifty two adolescents undergoing 3rd molar surgery participated. Adolescents' negative affectivity, expectancies, coping styles and illness behavior encouragement were assessed before surgery. Extent of surgery was assessed by the oral surgeon. Outcome measures included resumption of initial mouth opening, disability and pain.

After controlling for extent of surgery, the psychosocial parameters accounted for 16% of the variance in mouth opening and 24% of the variance in disability. Extent of surgery did not predict recovery. Expectancies had an independent and significant association with recovery. Gender effects masked associations between some predictors and recovery. Adding pain as a predictor more than doubled the variance explained in disability. Expectancies about recovery should be included in models of adolescents' recovery from surgery. Interventions which modify pessimistic expectancies and improve pain treatment may enhance recovery.

T8. Patricia A. McMullen, John D. Fisk*, and Stephen J. Phillips**
* Camp Hill Medical Centre
** Department of Medicine, Dalhousie Univeristy

FACES ARE SPECIAL: AN UNUSUAL CASE OF VISUAL AGNOSIA

A long-standing issue in visual cognition is whether face recognition relies on the same processes as object recognition. Visual agnosics are brain damaged individuals who have lost the ability to recognize objects through vision but still successfully recognize through other modalities such as touch. Face recognition can be lost with spared object recognition, but to date few have reported lost object recognition with spared face recognition. Patient H.H. showed dramatic deficits in the ability to recognize simple objects, like letters and geometric shapes. More importantly, he failed to identify 14/15 real, common objects presented visually. However, his identification of these objects through touch increased to 12/15 correct. In sharp contrast, H.H. correctly identified 6/8 coloured, magazine photographs of famous people and successfully sorted (20/22) line-drawings of faces with the features in correct spatial position from those in which the features were spatially scrambled. These results support the notion that face processes are unique and as such, not shared by the object processing system.

T9. Natalie A. Phillips and Jeannette McGlone

FEW COGNITIVE ABILITIES CHANGE FOLLOWING TEMPORAL LOBECTOMY

The frequency of clinically significant change greater than one standard deviation on 14 cognitive neuropsychological tests was examined in 47 patients undergoing temporal lobectomy for medically intractable epilepsy. The majority of scores (68%) did not change more than one standard deviation, and 17% actually improved following surgery. Clinically significant decline in neuropsychological functioning was relatively infrequent (15%). Data were also grouped into material-specific domains (verbal versus non-verbal memory, and language versus visuospatial abilities) to determine how often decline was related to side of surgery in 35 right-handed patients (13 left-temporal; 22 right-temporal). Material-specific change occurred only after left temporal lobectomy where relatively more verbal memory scores declined (17%) than improved (8%) and more non-verbal memory scores improved (22%) than declined (3%). Taken together, these data indicate that most patients with longstanding temporal lobe seizures experience relatively little cognitive change following a temporal lobectomy.

T10. Bruno J. Losier, Jeannette McGlone and Susan E. Black[1]

NO SEX DIFFERENCES IN THE INCIDENCE OR SEVERITY OF LEFT NEGLECT AFTER UNILATERAL STROKE

One hundred and thirty-eight patients with unilateral stroke were administered a bedside battery for hemispatial neglect and the Visual Search task (Kimura, 1986). All were pre-morbidly right-handed, and free of prior neurological and psychiatric histories.

Chi-square analyses revealed a higher incidence of left neglect following right than left-hemispheric lesions, but suggested no differences between men and women. Analyses of variance (Side of Lesion, Sex) on the composite score of the bedside battery and search time on the Visual Search Board, demonstrated that the severity of neglect was greater following right- than left-hemisphere damage and revealed longer search time for objects located in the contralesional half of the page than for those appearing on the ipsilesional side, respectively. Again, no significant differences between men and women were found in either analysis.

Although the incidence and severity of neglect was greater following right-hemisphere damage, the present study provided no support for sex differences in the manifestation of hemispatial neglect.

[1]Cognitive Neurology Division, Sunnybrook Hospital, University of Toronto Centre, Toronto, Ontario.

T11. Farhad N. Dastur and Richard E. Brown

THE DEVELOPMENTAL EFFECTS OF THYROID HORMONES IN LONG-EVANS RAT PUPS

Sixty Long-Evans rat pups were subcutaneously injected with l-3,5,3'-triiodothyronine (l-T3) from postnatal days two to five. Twelve pups (6 male and 6 female) were randomly assigned to each of five conditions: 0.1, 0.5, or 1.0 µg/g b.w., a saline control group (equivalent volume), or a non-handled control group. The effects of l-T3 were measured for each group from postnatal day 8 to 21 using a 14 item developmental test battery designed to measure alterations in the timing and form of simple reflexes, sensory function, motor coordination, homeostatic regulation, and growth. The tests included measurements of ultrasonic vocalizations at multiple frequencies, thermoregulation, grooming, general activity, tremors, eye opening, ear opening, surface righting, negative geotaxis, acoustic startle reflex, grasping reflex, placing reflex, brain weight, and body weight. In experiment 2, the same battery was used to assess the development of pups made hypothyroid by administration of the goitrogen methimazole (MMI; 0.1 mg/ml) in the drinking water of dams on gestational day 17 through postnatal day 10. Preliminary results suggest that l-T3 accelerates development while MMI retards development. Neither l-T3 nor MMI appeared to alter the form of development.

T12. Hugh Piggins and Benjamin Rusak

GASTRIN-RELEASING PEPTIDE (GRP1-27): WHAT'S A NICE PEPTIDE LIKE YOU DOING IN A PLACE LIKE THIS?

Gastrin-Releasing Peptide (GRP1-27) is a 27 amino acid peptide found throughout the mammalian central and peripheral nervous systems. GRP1-27 is found in high concentrations in the suprachiasmatic nucleus (SCN) of the hypothalamus which houses the circadian pacemaker. To date few studies have address the possible role of GRP1-27 in circadian pacemaker activity. Data from ongoing electro-physiological and behavioural experiments in our laboratory indicate that exogenous applications of GRP1-27 and structurally related peptides can activate SCN cells and phase-shift the circadian pacemaker. The data from these studies raise the possibility that GRP1-27 participates in photic entrainment processes.

T13. Dennis Phillips

ON HOW LONG IT IS AND WHETHER ANYONE CAN TELL THE DIFFERENCE

There is a claim, and from not disreputable sources, that the auditory perceptual system parses incoming information into chunks with a minimum duration of about a hundred milliseconds. A corollary of this view is that the minimum duration of an auditory percept is also about a hundred milliseconds (Efron, 1970). Presumably, this means that normal subjects are unable to discriminate the durations of auditory events that are less than that critical value, and there is some evidence that this might be the case. We have probed this notion by having subjects perform a discrimination of the duration of a temporal gap within wideband noise stimuli, in a 2AFC paradigm that tracks the subject's threshold (Levitt, 1970). Our studies indicate that subjects have little difficulty discriminating the duration of temporal gaps as short as 3ms and as long as 100ms, and with a resolution of about 15% of the reference duration. We are now studying the same subjects, with the same general paradigm, but for their ability to discriminate the duration of noise pulses (rather than the duration of a gap in the noise). Our experiments reveal that this task is even easier for the subjects. These data suggest that if the auditory perceptual system temporally "chunks" the incoming information, then the chunks are much shorter than 100 ms in duration.

T14. Stephen R. Shaw

RE-EVALUATION OF THE MOST SENSITIVE KNOWN "VIBRATION" DETECTOR

In vertebrate auditory systems, basilar membrane vibration has been measured directly: absolute threshold is reached at displacement amplitudes of ~4 x 10^{-9} meter, i.e. at dimensions that are small on an atomic scale. It has been claimed that the supersensitive knees of some insects do even better by the large factor of ~200 (0.002 nm threshold), raising the question of whether these magnificent animals have discovered some superior biophysical principle that has been missed in vertebrate evolution. The lab cockroach is the best of all, but previous students of its knees have failed to agree on just how sensitive these really are. Further, none of them calibrated their vibrators directly at the tiny amplitudes with which they worked, but trusted the perilous procedure of unsupported linear extrapolation downwards from much larger calibrated movements. Finally, some investigators delivered stimuli that may have unwittingly included an acoustic component. We have subsequently shown that the cockroach leg's "vibration" detector is a sensitive auditory organ, and that this may even be its primary response mode. In light of this, a new device has been developed that allows the leg's vibration threshold to be measured directly without acoustic complications. Overlapping calibrations eliminate the potential problems issuing from non-linear extrapolation. Some preliminary results will be described.

T15. Andrew Stuart and Dennis Phillips

TEMPORAL RESOLUTION IN DIFFERENT FREQUENCY CHANNELS OF HUMAN COCHLEAR OUTPUT

Last year, Phillips and colleagues reported that patients with selective high-frequency cochlear hearing losses evidenced a temporal resolution defect in their auditory perception. This defect expressed itself as a compromised ability to discriminate spoken words against a background of noise that fluctuated in amplitude, a task that requires the auditory system to resolve the speech fragments that occur in the troughs of the noise envelope. We argued that these listeners were performing the task using their (normal) low-frequency hearing, and that the low frequency channels of cochlear output have inherently poor temporal resolution. To test this hypothesis we have studied normal listeners with the same general paradigm, and under two listening conditions: (a), normal, and (b), one which prevented them from using their high-frequency hearing, namely low-pass filtering of the stimuli. The pattern of responses obtained under the "filtered-stimulus" condition precisely mimicked those obtained in patients with high-frequency, sensorineural hearing loss. This made us very happy troopers.

T16. Benjamin Rusak and Hiroshi Abe

CIRCUITOUS CIRCADIAN CIRCUITS

The suprachiasmatic nucleus (SCN) of the hypothalamus is the major pacemaker of the mammalian circadian timing system. It is synchronizedby environmental light cues conveyed from the eyes to the SCN directly via the retinohypothalamic tract and indirectly through a portion of the lateral geniculate nuclei called the intergeniculate leaflet (IGL). We have used light-induced expression of the immediate-early gene c-*fos* as a marker for retinal input to the SCN, in order to study which neurotransmitter receptors must be activated in order to convey light information to this nucleus. Evidence from many sources indicates that the NMDA receptor is critical to the photic entrainment process. Nevertheless, neurons in one restricted portion of the SCN continue stubbornly to express c-*fos* under photic control, despite blockade of either NMDA or non-NMDA amino acid receptors. We found evidence that suggested that these cells were continuing to get input indirectly from the IGL, but further tests have refuted this model. Our newer results now suggest that there are two pharmacologically and anatomically distinct direct retinal projections to the SCN. This revised model may also require reinterpretation of the results of several earlier studies.

T17. Gail Eskes

THE ROLE OF THE PREFRONTAL CORTEX IN CONDITIONAL DISCRIMINATION LEARNING

Rats with prefrontal lesions are impaired on conditional discrimination learning (CDL), a task that requires them to associate one response to a particular stimulus and a differentresponse to another stimulus. The role of the prefrontal lobe in this task is unclear. Deficits on CDL could be due to either an inability to learn the necessary associations between the different stimuli and responses, and/or to a difficulty in selecting the correct response, given a particular stimulus. The present study was designed to attempt to distinguish between these two alternatives. Rats with prefrontal lesions (PFC, n=10) and sham-operated controls (SC, n=11) were first compared using the standard CDL paradigm, in which they were required to learn to respond to one of two bars, depending on the location of the light stimulus. We confirmed the impairment of PFC lesioned animals on this paradigm as previously reported (Winocur, 1991). In the next phase, we attempted to eliminate the necessity of selecting between the two equally probable response alternatives by providing only one bar during any one light stimulus. Correct responding (either pressing the correct bar, or not pressing the incorrect bar) was rewarded. Thus, in this new procedure, rats were tested for their knowledge of the association between the light and bar, in the absence of the need to make a choice between the two available bar press responses. Under these conditions, PFC animals were not impaired on CDL when compared to controls. Performance averaged over the last 2 days of testing were: 67.4%± 4.39 correct vs 66.49% ± 4.53, PFC vs controls, respectively). These data suggest that deficits seen in conditional discrimination learning due to prefrontal lesions are due to a difficulty in selecting between response alternatives, rather than in forming associations between stimuli and responses.
(In collaboration with Gordon Winocur)

T18. Tara Fidler and Vincent M. LoLordo

EFFECTS OF PREEXPOSURE ON FEAR CONDITIONING TO A CONTEXT

Nonreinforced exposure to a stimulus typically makes it harder to condition that stimulus when it is reinforced. This phenomenon is called latent inhibition. In the present experiment, which used a 3 X 2 factorial design, groups of rats were repeatedly, but briefly, preexposed to: (1) the context in which they would subsequently be conditioned, or (2) a tone in that context, or (3) a different context. Then rats from each of these groups were randomly assigned to two subgroups. One set of subgroups received several sessions in which the tone occurred and was followed by a brief footshock. The others received the shock but not the tone. Conditioned freezing was assessed. Rats that were preexposed to the tone in the conditioning context and then given tone-shock pairings in that context learned to freeze in that context sooner than any of the other groups, and maintained a higher level of freezing there over a number of extinction sessions. This result is noteworthy because it is the opposite of the latent inhibition usually obtained. We will speculate about its basis.

T19. Werner K. Honig

NUMEROSITY DISCRIMINATIONS BY PIGEONS OBTAINED WITH A TOUCH-SCREEN PROCEDURE

The "touch screen" provides a sensitive method for assessing the discrimination of complex arrays of elements. Pigeons were trained to peck at one side of a randomized array when there were more blue than red elements, and at the other side with the opposite proportions. This yielded orderly maintained discriminations of response location based on relative numerosity. Corresponding results were obtained when pigeons had to respond to the left side when uniform arrays were presented, and to the right side when other, mixed proportions were shown. This general method has the advantage that the different elements are equally associated with reinforcement during training.

T20. Kelly A.K. Forbes and John F. Connolly

MEMORY PERFORMANCE, ERPS AND PHONOLOGICAL ENCODING DURING SILENT READING

The dual-route theory (e.g. Coltheart, 1978) proposes that the initial interpretation of written text is phonologically mediated. Phonological mediation is later superseded by orthography as reading becomes more proficient. Research indicates that the interpretation of written text may continue to be influenced by the phonological representation of the words (e.g. Doctor & Coltheart, 1980). The present study investigates the effects of prior exposure on the later interpretation of written text containing homophones using explicit and implicit memory measures. Subjects were right-handed, English speaking undergraduates (N=40). Event related potentials (ERPs) were recorded to highly contextually constrained sentences completed with homophones. The manner in which the sentences were completed was manipulated in the study phase. Sentence endings were completed with homophones that were semantically and phonologically appropriate (S^+P^+), semantically incongruous but phonologically appropriate (S^-P^+) or semantically and phonologically incongruous (S^-P^-) to the sentence context. During the test phase, subjects completed a recognition task or an oral spelling task requiring them to identify homophones previously seen or to spell homophones the first way they came to mind, respectively. ERP findings indicate that S^+P^+ and S^-P^+ are processed similarly and both are processed differently than S^-P^-. In contrast, behavioural measures indicate that comprehension is influenced by spelling frequency and the degree to which the information agrees with the sentence context. Findings support phonological mediation in reading and are discussed in light of current theories.

P1. Kelly Frame, John Christie and Raymond Klein

NEGATIVE PRIMING IN A LOCALIZATION TASK: A METHODOLOGICAL CRITIQUE AND AN EMPIRICAL PUZZLE

When a subject is presented with pairs of trials (prime followed by probe) in which a target (O) must be localized, sometimes in the presence of a distracting non-target (+, distractor), response time is slower if a probe trial's target appears in the location occupied by the previous (prime) trial's distractor than when it appears in a new location. This finding (negative priming) supports the proposal that selective attention is accomplished by active inhibition of irrelevant information. The literature, however, is characterized by an insensitivity to design features necessary to ensure the purity of the observed effects. When all necessary conditions are included, our results reveal interference, negative priming, and both inhibition and facilitation at the previous target's location. Yet these mechanisms, alone, are insufficient to explain the data. Will there still be a role for negative priming when the full puzzle is solved?

P2. Jeff P. Hamm and Patricia A. McMullen

RESPONSE COMPETITION CONTRIBUTES TO THE EFFECTS OF ORIENTATION WHEN LOCATING THE TOP OR BOTTOM OF ROTATED OBJECTS

The time to determine if a dot is located near the top or bottom pole of rotated objects increases linearly for rotations up to 120 deg. This effect has been linked to a similar effect of orientation found for rotated object naming. However, different causes for these effects may be inferred if it was shown that they were partly due to response compatibility in the top-bottom discrimination task. Results from this task showed that responses to dots at the top were faster than those to dots at the bottom of the screen. This effect was additive with the effects of object orientation. However, compatibility between location of the dot on the screen and response was confounded with object orientation. Objects near upright orientations were confounded with compatible responses and objects at more extreme orientations were confounded with incompatible responses. To determine if response compatibility accounted for some of the effects of orientation during top-bottom discriminations, "top" and "bottom" keys were positioned horizontally or vertically. Vertical key placement, which was thought to improve response mapping, resulted in a greater effect of orientation than horizontal placement. The effects of orientation during top-bottom discriminations may partially reflect response competition, suggesting these orientation effects are not reflective of those found during rotated object naming.

P3. Christopher Dywan, Joseph Byrnes and John Connolly

ELECTROPHYSIOLOGICAL CORRELATES OF SINGLE-WORD RECEPTIVE VOCABULARY IN CHILDREN

A computerized version of the Peabody Picture Vocabulary Test, Revised, was developed to examine potential electrophysiological correlates of single-word receptive vocabulary in normal 10-year-old children. Of primary interest will be ERP components that may differentiate between picture-word pairs that match or mismatch at 3 levels of difficulty (below age, at age and above age level). This paradigm may potentially facilitate objective assessment of single word receptive vocabulary in patients whose conditions, such as Cerebral Palsy, may preclude traditional assessments based on verbal or motor responses. If available, preliminary data will be presented and discussed.

P4. Heather A. Schellinck and Patricia A. McMullen

WHY DOES ORIENTATION-SENSITIVITY DIMINISH DURING REPEATED NAMING OF ROTATED OBJECTS: FEATURES OR OBJECT-CENTRED REPRESENTATIONS?

Effects of orientation on rotated object naming diminish over presentation blocks. Is this effect due to emergent naming based on orientation-invariant features (Jolicoeur & Milliken, 1989) or on object-centred representations (see Marr, 1982)? If the former, then repeated naming of objects whose features are spatially scrambled should diminish effects of orientation on a final block of intact, rotated object naming in the same way that repeated blocks of intact, rotated object naming does. In contrast, if naming over blocks is based on emergent reliance on object-centred representations, then repeated naming of feature-scrambled objects should not result in diminished effects of orientation on a final block of intact, rotated object naming. To test which hypothesis is better supported, one group of subjects repeatedly named feature-scrambled objects followed by a final block of intact, rotated object naming. A second group named intact rotated objects in all blocks. A final group repeatedly named intact, upright objects followed by a block of intact, rotated objects. Effects of orientation on the final block, were diminished after repeated naming of intact, rotated and feature-scrambled objects but not after repeated naming of upright objects. These results support the hypothesis that feature-based naming underlies the diminished effects of orientation over repeated blocks of rotated object naming.

P5. Peter C. Kind, Chris J. Beaver and Donald E. Mitchell

THE EFFECTS OF MONOCULAR DEPRIVATION AND REVERSE OCCLUSION ON THE EXPRESSION OF CAT-301 IN AREA 17 OF KITTENS

Although considerable recovery can occur from the effects of an early period of monocular deprivation (MD) following prompt imposition of a period of reverse occlusion (RO), recent behavioural work has revealed that this procedure can result in a wide variety of other outcomes, including severe bilateral amblyopia. In an attempt to understand the mechanisms that underlie the behavioral findings, we have examined the expression of the Cat-301 antigen in area 17 of kittens monocularly deprived to 4, 5, 6 or 8 weeks of age that subsequently received either a period of RO (that lasted from 9 days to 12 weeks) followed by binocular visual exposure, or else the latter alone. We have reported previously that certain of these regimens that employ short periods of RO result in very low levels of Cat-301 expression in the dLGN, equivalent to levels reported in dark-reared animals. We now report that these same rearing regimens also result in a substantial reduction of Cat-301 expression in area 17. The largest decrease (as much as 75%) occurred in the supragranular layers with a smaller decrement (about 30%) evident in the upper portion of layer 4.

P6. Judith Francis and Karina Davidson

IN THE MOOD: INFLUENCE OF MOODS ON SELF-DISCREPANCIES

The purpose of the present study was to examine the causal relation between self-discrepancies and mood. According to Self-discrepancy theory (e.g. Higgins, 1987), various self-concept discrepancies represent enduring cognitive structures, which when activated produce specific changes in mood. The purpose of the present study was to test the hypothesis that transient mood states can influence the type of self-discrepancies reported.

Subjects were assigned to one of two mood induction conditions: sad, or anxious. After mood was induced, subjects completed the Selves Questionnaire which measures self-concept discrepancies. Results of analyses support Higgins' proposed causal relation between mood and self-discrepancies. When subjects were classified as dysphoric or non-dysphoric based on BDI scores, dysphoric subjects were found to have higher ideal discrepancy scores as well as greater ought discrepancy scores, regardless of mood induction condition. Results support a uni-directional relationship between self-discrepancies and mood and argue against a reverse causality hypothesis.

P7. Kelly A. Frame

EFFECTS OF SEX HORMONES ON COGNITIVE ABILITIES: A CRITICAL REVIEW

Effects of sex hormones on cognitive abilities in humans has been investigated 1) prenatally, by acting to organize neural pathways and 2) by activating previously organized functions in adulthood. Because it is difficult to manipulate hormones in humans, research on activational effects has relied upon naturally occurring bio-rhythms, and medical conditions. The menstrual cycle is a convenient way to study how hormonal fluctuations might influence cognitive abilities in women. For example, high levels of estrogen (follicular phase) have been associated with poorer performance on spatial tasks compared to performance during menses when hormones are low (Hampson, 1990). Other studies have focused on a cycling of cognitive performance in relation to brain laterality effects. Heister et al. (1989) reported right hemisphere superiority on a visual perception task during menses which decreased linearly through follicular and luteal phases to a small left hemisphere superiority effect during the premenstrual phase. To date, theoretical explanations are unconfirmed and methodology is problematic. A summary of relevant results as well as theoretical implications and methodological suggestions will be proposed.

P8. Michael MacGregor and Karina Davidson

SEX DIFFERENCES IN TARGET HOSTILITY RATINGS

This study examined: 1) Sex differences in the ability to detect hostility during an audio recording, and; 2) the extent to which such differences affect subject- and friend- agreement about the subject's hostility level. Data was collected from 112 undergraduates who each provided the name of a close friend. To determine their ability to detect hostility the friends completed the MacGregor Hostility Recognition Scale (MACH-I) after listening to either a high or low hostile audio recording. Results indicated that in both the high and low hostile recordings males (M=3.4 and M=2.7 respectively) reported significantly less hostility to be present than females (M=4.0, t(52)=-2.9,p_.005 and M=3.1, t(51)=-1.8, p_.056 respectively). When male and female ratings of hostility were subtracted from the actual levels of hostility in the recordings (as determined by expert raters) females (Mdiff=.27) were significantly more accurate than males (Mdiff=.86, t(52)=2.9, p_.005) on the high hostile recordings. No difference in accuracy was found between males (Mdiff=-.72) and females (Mdiff=-1.13, t=1.80, p>.05) on the low hostile recordings.

Second, the subject's hostility level, measured by the Cook-Medley Hostility Scale, was rated by both the subject and the subject's friend and subject-friend agreement was determined. Male friends had less agreement with male subjects about hostility (M=19.43, SD=10.14) compared to female friends with male subjects (M=25.47, SD=8.30, t(34)=-2.00, p_.053). Thus it appears that males are less accurate in detecting hostility.

P9. Rita Ingenmey and Philip J. Dunham

GENDER DIFFERENCES IN SOCIAL REFERENCING DURING A STRANGER INTERACTION

We asked 1) whether 9-month-old males and females differ in their tendency to visually reference their mother during the infants' interaction with a stranger and 2) if gender differences are observed in infant's emotional reactions. Maternal employment conditions were also included as a moderator variable. Thirty-four infants played "peek-a-boo" with a female experimenter while the mother's profile was in the infant's view. Looks at mother were used as an index of social referencing and Izaard's (1983) AFFEX was used to assess the infant's emotional expression. Females referenced their mothers more $F(1, 29) = 6.05$, $p < .02$, than males but no gender differences were observed in the infants' emotional reactions. Infants with mothers employed outside the home smiled more frequently at the stranger, $F(1,29) = 7.58$, $p < .01$, than infants with mothers working inside the home. The results suggest that gender differences in social referencing behaviour are observed as early as 9 months and that these gender differences in referencing are not emotionally mediated.

P10. Peter Hall and Karina Davidson

GENDER DIFFERENCES IN THE CONSTRUCT VALIDITY OF TWO HOSTILITY MEASURES

Objective: To determine the construct validity of the Ho subscale of the MMPI and the structured interview method (SI) of hostility measurement while examining gender differences.

Design: Correlational

Setting: University

Participants: From an introductory psychology class. Mean age of nineteen. 54 male and 87 female volunteers were used.

Interventions: Subjects were given a standard structured interview and asked to fill out the Ho scale and the PANAS negative affect (NA) scale.

Measures: Two tailed t-tests were used to compare means of males with females. Simple correlations and intercorrelations between Ho and SI measures were used.

Results: For females 9% of the variance ($r=.30$) in scores on the Ho scale were accounted for by variability of scores on the NA scale. No significant variability in PH scores was accounted for by NA scores in females. In males, 11% of the variance ($r=.34$) in scores on the Ho scale were accounted for by variability in NA scores. Scores of PH accounted for an additional 33% of variability in Ho scores in males.

Conclusions: It appears that Ho and SI methods of hostility assessment are not measuring the same construct for females, but have high construct validity for males. Rater-reliability and other possible confounding factors for the female hostility data will be presented. There are significant gender differences in expression of hostility.

AWARDS

In the spirit of years past, the In-House Convention Committee would like to continue the fine tradition of presenting an award for the most entertaining title. However, this year we would like to put a twist into the tradition by asking you, the attendees, to chose via the democratic method, those titles that you feel are most deserving.

Please denote your choice in the spaces provided below, and place the ballot in the box located in the presentation room.

I cast my vote for the MOST ENTERTAINING/INTERESTING title to:

Title: ____

Author(s): ____

For those who do not equate "entertaining/interesting" with priapic proclamations, we provide the following:

I cast my vote for the MOST SALACIOUS title to:

Title: ____

Author(s): ____

The awards ceremony will take place on Friday afternoon during the closing remarks of the conference. Don't miss it!